

ORANGE, MA, USA

WMO: 725085

Lat: 42.570N

Lon: 72.291W

Elev: 169

StdP: 99.31

Time Zone: -5.00 (NAE)

Period: 96-19

WBAN: 54756

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.3	-16.3	-25.2	0.4	-16.9	-22.5	0.5	-14.4	10.4	-4.5	9.2	-4.3	1.2	290	0.519

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	32.0	22.0	30.2	21.1	28.7	20.3	23.7	28.6	22.9	27.4	22.1	26.3	4.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.3	17.4	25.7	21.6	16.6	25.0	20.9	15.9	24.4	71.5	28.7	68.3	27.6	65.6	26.0	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.9	7.9	7.0	-23.8	34.6	3.7	1.7	-26.5	35.8	-28.6	36.8	-30.7	37.7	-33.4	38.9		
(5)			-24.1	25.2	3.6	0.7	-26.7	25.7	-28.8	26.1	-30.8	26.5	-33.4	27.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.6	-5.2	-3.6	0.9	7.5	13.8	18.6	21.6	20.7	16.5	9.6	3.6	-1.5		
	DBStd	10.31	6.04	5.37	5.12	4.61	4.31	3.75	3.04	3.05	4.01	4.47	4.72	5.00		
	HDD10.0	1871	471	381	286	102	11	0	0	0	2	63	198	357		
	HDD18.3	3833	729	614	541	326	152	43	7	12	82	272	441	614		
	CDD10.0	1363	0	0	3	28	130	257	360	331	197	51	7	1		
	CDD18.3	283	0	0	0	1	12	49	108	84	27	1	0	0		
	CDH23.3	2800	0	0	4	38	211	512	1059	741	225	12	0	0		
	CDH26.7	865	0	0	1	12	63	163	365	206	56	1	0	0		
(14) Wind	WSAvg	2.5	2.6	2.8	3.1	3.0	2.6	2.3	2.1	1.9	1.9	2.3	2.4	2.5		
(15) Precipitation	PrecAvg	1199	91	78	99	95	97	120	99	104	106	109	97	107		
	PrecMax	1589	152	267	173	164	151	235	202	229	286	357	151	200		
	PrecMin	880	36	36	22	25	35	17	37	35	14	20	32	44		
	PrecStd	211	28	47	42	42	30	57	48	46	67	72	29	38		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.5	14.0	21.6	28.1	31.4	33.0	34.2	32.8	31.2	24.9	18.9	16.1		
		MCWB	11.2	9.8	13.4	16.1	18.9	22.0	23.7	22.9	22.1	17.5	15.1	13.2		
	2%	DB	8.1	9.7	15.4	22.8	28.3	30.9	32.1	30.8	28.0	22.2	16.8	11.4		
		MCWB	5.7	5.9	9.1	12.8	18.1	21.2	22.3	22.2	20.3	17.0	13.8	9.5		
	5%	DB	5.1	6.7	11.9	19.4	25.7	28.3	30.3	28.9	25.9	19.9	14.2	8.6		
		MCWB	2.9	3.4	6.9	11.7	16.3	19.7	21.4	20.9	19.2	16.0	11.3	6.7		
	10%	DB	2.7	3.8	8.7	16.9	22.6	26.4	28.7	27.4	23.8	17.4	12.0	6.0		
		MCWB	0.9	1.2	5.2	10.0	15.0	18.4	20.6	20.2	18.4	13.7	9.1	3.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.8	12.0	14.6	17.8	21.2	23.9	25.1	24.6	23.4	20.6	16.8	13.2		
		MCDB	12.3	13.2	20.1	23.8	28.1	29.4	31.9	30.0	27.5	23.1	18.1	14.3		
	2%	WB	6.4	6.6	10.7	15.4	19.6	22.7	23.8	23.6	22.1	18.5	14.4	10.1		
		MCDB	7.5	9.1	13.9	20.9	25.0	27.9	29.1	27.9	25.6	20.5	16.1	11.4		
	5%	WB	3.2	3.5	7.7	13.2	18.2	21.7	22.9	22.7	21.1	16.6	12.0	6.8		
		MCDB	4.7	6.0	10.7	17.1	23.0	26.2	27.6	26.6	24.2	19.1	13.8	8.2		
	10%	WB	1.2	1.7	5.4	10.9	16.6	20.4	22.1	21.8	19.8	14.5	9.3	4.0		
		MCDB	2.4	3.4	8.7	15.4	20.9	24.0	26.4	25.4	22.8	17.0	11.6	5.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.0	11.0	10.7	12.8	12.9	12.0	12.2	12.0	12.3	11.6	10.4	9.2		
		MCDBR	11.1	13.6	16.2	18.6	18.0	15.2	14.1	13.6	14.2	14.2	13.3	12.0		
	5% WB	MCWBR	9.4	10.3	10.2	9.6	8.4	6.7	5.6	5.9	7.3	8.8	10.1	10.4		
		MCWBR	10.6	12.3	14.1	15.2	14.4	12.5	12.0	10.7	11.1	11.8	11.6	11.2		
(40) Clear-Sky Solar Irradiance	taub	taub	0.284	0.294	0.321	0.366	0.399	0.442	0.460	0.429	0.374	0.332	0.303	0.283		
		taud	2.442	2.416	2.442	2.372	2.316	2.221	2.179	2.252	2.387	2.520	2.555	2.534		
	Ebn at Noon	Ebn at Noon	865	913	926	902	876	836	815	829	856	859	840	840		
		Edh at Noon	75	90	100	116	126	139	144	130	105	82	68	63		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.74	2.60	3.65	4.61	5.09	5.50	5.80	5.21	4.24	2.72	1.89	1.41		
	RadStd	RadStd	0.13	0.22	0.25	0.50	0.59	0.53	0.36	0.27	0.30	0.28	0.17	0.15		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+135	+57
(47) Regional (76 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+94	+42

Nomenclature: See separate page